Marmara University

Engineering Faculty, Electrical and Electronics Engineering

EE4077: Fundamentals of Machine Learning Midterm Exam Solutions, Spring 2025

1. (a) Observation matrix $X = \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{bmatrix}$ (2), target vector $\boldsymbol{y} = \begin{bmatrix} 2.5 \\ 4.3 \\ 5.5 \\ 6.1 \\ 6.1 \end{bmatrix}$ (2), parameter vector $\boldsymbol{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ (2)

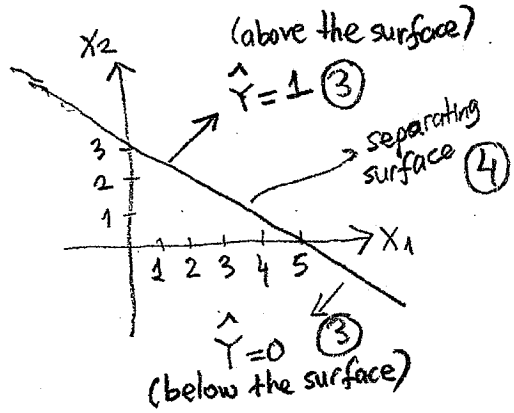
(b) $$w^* = \left(\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 9 & 16 & 25 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 4 \\ 3 & 9 \\ 4 & 16 \\ 5 & 25 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 9 & 16 & 25 \end{bmatrix} \begin{bmatrix} 2.5 \\ 4.3 \\ 5.5 \\ 6.1 \\ 6.1 \end{bmatrix} = \begin{bmatrix} 2.77 \\ -0.31 \end{bmatrix}$$ (3)

so the desired optimum model is $\hat{y} = 2.77x - 0.31x^2$ (3)

(c) The model is unique since there is only one solution. (4)

(d) We do not need to apply ridge regression since the error surface is convex (it has only one minimum) (4)

2. (a)



(3) (4) (3)

(b) Computing $P[Y = 1]$ gives

$$P[Y = 1] = \frac{1}{1 + e^{-(3\times3+5\times9-15)}} = \frac{1}{1 + e^{-39}} > 0.5$$ (5)

Since $P[Y = 1] > 0.5$, our prediction is $\hat{Y} = 1$ (5)

3. (a) The classifier is logistic regression, it makes 6 mistakes (4) (3)

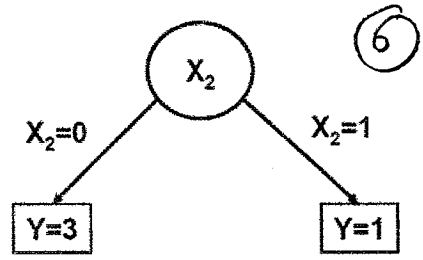
(b) The classifier is decision tree, it makes 2 mistakes. (4) (3)

(c) The classifier is k-nearest neighbor, it makes no mistakes.

(3) (3)

4. (a) When $X_1 = 0$, Y takes on 3 different values while $X_1 = 1$, Y takes on 2 different values. On the contrary, when $X_2 = 0$ or 1, Y takes on 2 different values. In other words, observing X_2 reduces the uncertainty about Y more. Hence, X_2 provides more information gain. (6)

(b) Since X_2 provides more information gain, we choose to split on X_2. The resulting decision tree is shown below (4)



(6)

(c) An example with $X_1 = 0$ and $X_2 = 1$ will be classified as $Y = 1$ on this three since $X_2 = 1$ (4)

5. (a) FALSE because taring error always decreases as model complexity increases (4)

(b) FALSE because validation error first decreases then it increases with respect to complexity (4)

(c) TRUE because a simple model has high bias and low variance (4)

(d) TRUE because a complex model has low bias and high variance (4)

(e) FALSE because a simple model underfits and a complex model overfits. (4)